

Total No. of Questions :6]

SEAT No. :

P61

OCT. -16/BE/Insem. - 115

[Total No. of Pages :2

**B.E. (Mechanical)
CAD/CAM & AUTOMATION
(2012 Pattern) (Semester - I)**

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer three questions.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicate full marks.*
- 4) Assume suitable data if necessary.*

Q1) a) A triangle with vertices A (8, 0), B (12, 0) and C (12, 3) has undergone reflection about line $y=2$. Find the concatenated transformation matrix and then find new coordinates of triangle ABC using this transformation matrix. **[6]**

b) What do you understand by Composite and Inverse Transformation?**[4]**

OR

Q2) Find the concatenated transformation matrix, if the transformations are performed as per the following sequence:

- a) Rotation through 45 in anticlockwise direction.
- b) Translation through +5 and -8 units along the X and Y directions respectively.
- c) Rotation through 60 in clockwise direction. What is the effect of above transformations on triangle having coordinates A (0, 0) B (10, 0) and C (0, 8)? **[10]**

Q3) a) What are advantages of Solid modeling by CAD systems? And state types of Curves in CAD? **[4]**

- b) Calculate points on Hermite Cubic spline curve at value of $u=0, 0.2, 0.4, 0.6,$ and 0.8 having the endpoints P0 [4,4] and P1 [8,5] the tangent vector for end are P0[5,6] and P1[10,7]. **[6]**

OR

P.T.O.

Q4) Explain following terms with respect to solid modeling **[10]**

- a) Sweep and Extrude.
- b) CSG and B-Rep.
- c) Constraint Based Modeling.

Q5) M.S Tapered rod of $E=2 \times 10^6$ N/sq.mm, and 600mm length with 650 and 350 sq.mm c/s area at the ends. One end is fixed and other is subjected to axial tensile load of 40KN. Model the rod with 3-equal stepped rod and solve for following by FEM: **[10]**

- a) Global stiffness matrix
- b) nodal displacements
- c) stresses in elements

OR

Q6) a) Discuss shape functions in 2D CST finite Element with sketch? **[4]**

- b) The 1-D element has node-1 and 2 of stresses 200N/sq.mm and 300N/sq.mm with total length of 200mm. Calculate shape functions with nodes 1 and 2 if stress is to be calculated at point 'P' within the element situated at 160mm from node-1 also calculate stress at point 'P'. **[6]**

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